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Smart Learning Environments

REVIEW

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A systematic review of literature reviews on artificial intelligence in education (AIED): a roadmap to a future research agenda

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Abstract

Despite the increased adoption of Artificial Intelligence in Education (AIED), several concerns are still associated with it. This has motivated researchers to conduct (systematic) reviews aiming at synthesizing the AIED findings in the literature. However, these AIED reviews are diversified in terms of focus, stakeholders, educational level and region, and so on. This has made the understanding of the overall landscape of AIED challenging. To address this research gap, this study proceeds one step forward by systematically meta-synthesizing the AIED literature reviews. Specifically, 143 literature reviews were included and analyzed according to the technology-based learning model. It is worth noting that most of the AIED research has been from China and the U.S. Additionally, when discussing AIED, strong focus was on higher education, where less attention is paid to special education. The results also reveal that AI is used mostly to support teachers and students in education with less focus on other educational stakeholders (e.g. school leaders or administrators). The study provides a possible roadmap for future research agenda on AIED, facilitating the implementation of effective and safe AIED.

Keywords: Artificial intelligence, Generative AI, Education, Smart learning, Literature review, Meta-synthesis, Future research

Introduction

AI in education

The roots of Artificial Intelligence (AI) go back to the 1950s when John McCarthy organized a workshop at Dartmouth College aiming at defining AI (Russel & Norvig, 2010). The concept of AI was then widely and broadly discussed and investigated in many fields and domains, especially education. The growing usage of AI is reshaping the educational environment extensively in the last five years (Allam et al., 2023; Chu & Yang, 2022), as Artificial intelligence in education (AIED) has changed the education system, knowledge-sharing approaches to learning, cognition and development of civilization (Kaur,

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2021). Bojorquez and Vega (2023) highlight that AIED could advance learning practices, support teachers, and create more personalized learning opportunities for learners.

Despite the reported advantages of AIED, several researchers have indicated that the design of AIED environments and settings could be challenged in terms of the characteristics of learners and teachers, collaborative, social, affective, and meta-cognitive aspects of learning and this sometimes leads to unfair and unreliable outcomes (Luckin, & Holmes, 2016; Wang et al., 2024; Bhimdiwala et al., 2021). Moreover, Hwang et al. (2020) argue that the interdisciplinary nature of AIED could present challenges for researchers with different disciplinary backgrounds. Pinkwart (2016) reveals that AIED might face various future challenges such as intercultural and global dimensions, practical impact, privacy, interaction methods, collaboration at scale, effectiveness in multiple domains, and the role of AIED in educational technology. Thus, international organizations like UNESCO (2024) discuss the opportunities and risks of AIED as it faces linguistic, cultural, generational, and geopolitical challenges. Recently, major tech companies and institutions like Amazon, Google, Facebook, Knewton, and Carnegie Learning have spent millions on AIED research and development (Vallanc & McMahon, 2023).

There is an ongoing debate over implementing effective and safe AIED. Multiple researchers present systematic reviews to summarize past research to address the strengths and weakness of AIED, and summarize the findings, in order to enable accurate decision-making and provide comprehensive insights into AI use in education (Ouyang et al., 2023; Wang et al., 2024; Tlili, 2024a).

Research gap and study objectives

Literature reviews are a core research approach in developing new theories and identifying trends and gaps. Further, literature reviews occupy the top of scientific evidence and have an authoritative power. In that, when a systematic review concludes an intervention or a technique is effective, adoption follows on both individual and institutional levels. Therefore, analyzing systematic reviews is valuable in offering a synthesized summary of the worth or the state of the art of a field. Given the importance of AI in education, several systematic reviews have been conducted targeting several foci.

For instance, Wang et al. (2024) conducted a systematic literature review on AI in education based on primary categories of AIED applications, the research topics, guiding theories of AIED, methodologies, and research contexts. In a similar context, Bond et al. (2024) conducted a meta systematic review with a focus restricted to the last 5 years and higher education only. Boussouf et al. (2024) systematically reviewed AIED from the relationship between education and AI, their interaction, AI's contribution to educational evolution, and research challenges. Tlili (2024a) conducted a meta-analysis to investigate the effect of AI on learning achievement in computer science education, where he found that several variables might moderate this effect.

While these reviews and similar others have provided valuable insights into different perspectives, traditions and methodologies within the field, they also highlight the complexity and diversity of approaches. Additionally, they reveal a fragmented landscape that requires an overarching perspective. Therefore, it is evident that there is a need for a grand perspective that synthesizes the scattered evidence and provides researchers with synthesis of the existing evidence across time, disciplines and educational levels.

Furthermore, there is a need for a wider discussion on what it takes to design and implement effective AIED. With such disparity, a logical and appropriate next step is to conduct a systematic review of literature reviews, allowing the findings of separate reviews to be compared and contrasted, thereby providing comprehensive and in-depth findings for different stakeholders (Smith et al., 2011). As the development of AIED is rapidly advancing, this study conducts a systematic review of existing literature reviews on AIED that is neither restricted by time nor by educational levels. In doing so, we offer a comprehensive coverage that allows us to map the state of affairs of the AIED.

Through a systematic examination of high-quality published literature review articles, this study reveals the reported AIED trends and challenges, as well as the research gaps and future paths. These findings could help different stakeholders (e.g., policy makers, teachers, instructional designers, etc.) to facilitate the design and adoption of AIED worldwide. Although several systematic reviews of literature reviews have been conducted on different AIED subtopics, including blended learning (Ashraf et al., 2021), engineering (Kitchenham & Brereton, 2013), healthcare (Marcolino et al., 2018) and tourism (Pahlevan-Sharif et al., 2019), only two studies—Tlili et al., (2023a, 2023b) and Bond et al. (2024)—have undertaken a second-order approach (meta-review) as in our study. The meta-review by Tlili et al., (2023a, 2023b) focused exclusively on the transparency of the AIED systematic reviews rather than on the content of the AIED research. On the other hand, the meta-review by Bond et al. (2024) limited its scope to higher education and its temporal range to the last 5 years. Therefore, the presented results cannot be considered comprehensive of AIED as a whole. In the present study, we aim to capture the full breadth of the AIED research field by analyzing all of the existing reviews of AIED at any time point and within any subject area and educational level. Guided by the technology-based learning model (Hsu et al., 2012), this study aims to answer the following seven research questions (RQs):

- RQ1. What are the trends of AIED literature reviews in terms of publication year, geographic distribution, and publication venue?
- RQ2. Which stakeholders and educational levels are represented in AIED research?
- RQ3. What are the subject areas covered in AIED research?
- RQ4. What are the most frequently used research methods to review AIED research?
- RQ5. What are the targeted educational objectives in AIED research?
- RQ6. How was AIED designed in terms of the used techniques and technologies?
- RQ7. What are the advantages and challenges of AIED highlighted in the existing literature reviews?

Method

This study follows the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, that provide a standard peer-accepted methodology that uses a guideline checklist, which was strictly followed for this study, to contribute to the quality assurance of the revision process and to ensure its replicability (Page et al., 2021). In this context, several studies relied on PRISMA to conduct their systematic reviews (e.g., Zawacki-Richter et al., 2019; Zheng et al., 2023).

Theoretical framework

This study relies on the technology-enhanced learning model (Hsu et al., 2012) to analyze the reviewed studies and better understand AIED. This model presents several interconnected dimensions that should be looked at when applying a given technology in a learning environment (Lin & Hwang, 2019). The model has been used to analyze various educational technologies (Yang et al., 2021a, 2021b) and is based on six factors, namely subject area, learning models, participants, outcomes and issues, research methods and adopted technologies. The current study adopted most of the schemes from this model but made slight adjustments according to the AIED features, as depicted in Fig. 1.

Search strategy

To find all the relevant literature reviews on AIED, an extensive search was undertaken in relevant and well-known EdTech databases, namely Web of Science, Scopus, Taylor and Francis, IEEE Xplore, and ScienceDirect. To identify high-quality reviews and ensure reliable results, this study covered journal papers only (Tlili et al., 2024; Zawacki-Richter et al., 2019; Zheng et al., 2023). The following search string was used:

Search string: ((*AI substring*) AND (*Education substring*) AND (*Review substring*)).

- *AI substring:* "AI" OR "artificial intelligence" OR "machine learning" OR "deep learning" OR "natural language processing" OR "intelligent tutoring system*" OR "reinforcement learning" OR "generative artificial intelligence" OR "generative AI" OR "large language model*"
- *Education substring*: "education*" OR "tutor*" OR "teach*" OR "instruction*"
- *Review substring*: "literature review" OR "systematic review" OR "meta-analysis" OR "state-of-art"

After searching the relevant databases, two of the authors independently screened the retrieved papers' titles and abstracts, and excluded those that were clearly not literature reviews, such as empirical, descriptive and conceptual papers. The two authors then independently performed an eligibility assessment by carefully screening the full texts of the remaining papers, based on the inclusion and exclusion criteria described in Table 1.

The search process yielded a total of 1652 articles. After removing duplicates, 1565 papers remained. Then, 1109 papers were removed based on the screening of titles and abstracts. The remaining 456 papers were considered and assessed for full texts. 313 of

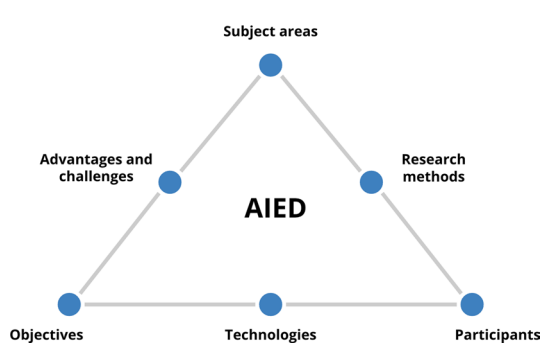

Fig. 1 Technology-enhanced learning model applied to AIED

Table 1 Inclusion and exclusion criteria

Inclusion	Exclusion
Journal article	Conference proceeding, book series, book chapters, books and dissertations
Literature review	Not a literature review or papers that do not give details about the way of conducting the literature review (e.g., method)
Focus only on AIED	Articles that focus on educational technology generally
Full text available	Full text not available
Articles written in English	Articles in languages other than English

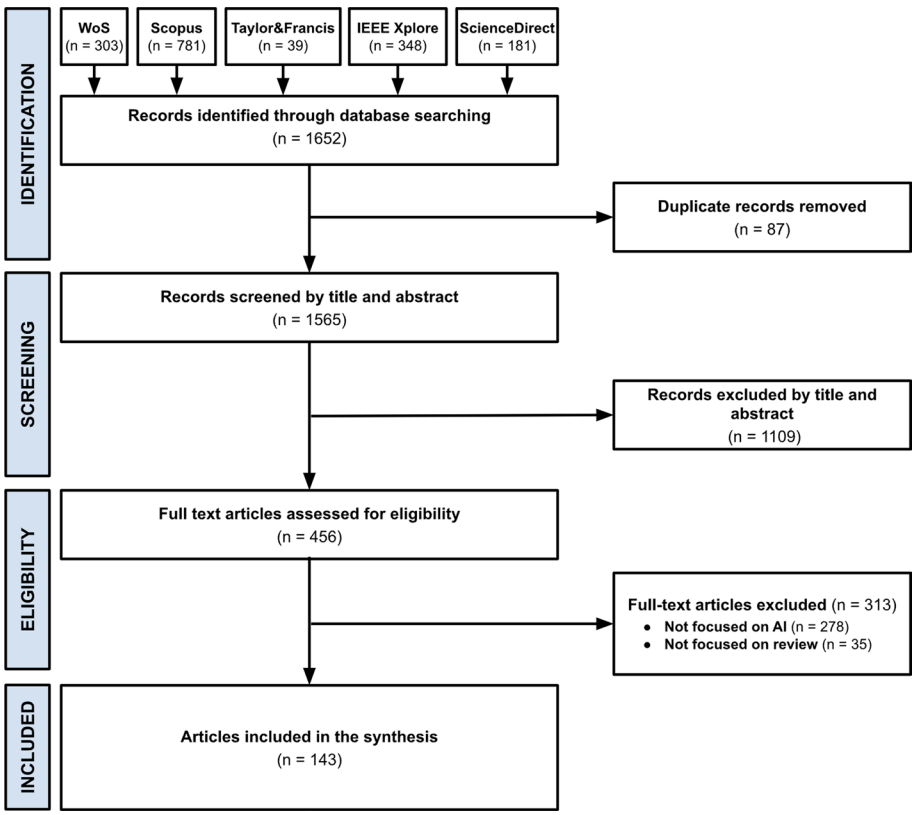

Fig. 2 PRISMA flowchart for data selection

these papers did not meet the inclusion criteria. Thus, as a total number, 143 eligible research studies remained for inclusion in the systematic review. Figure 2 presents the study selection according to PRISMA guidelines.

Data coding

The coding scheme was developed based on the Technology-enhanced learning model (see Fig. 1). Specifically, Table 2 presents a detailed description of the coding scheme that was used in this study to answer the aforementioned research questions. Weekly meetings were organized between the two coders during the coding process. During this phase, disagreements between the coders were resolved by discussion.

Table 2 The coding scheme

Item	Description	Coding
Year of publication	Year of publication	Year of publication
Country distribution	Country of the first author	Country
Journal	Journal name	Journal name
Subject area	The area of the conducted review (eg, mathematics, medicine or engineering)	The scheme of subject area refers to the general area where the study was conducted
Research methods	Whether the review is a systematic review or a meta-analysis	Since this paper is a literature review of literature reviews, the scheme of research methods mainly refers to the type of literature review including systematic review and metaanalysis review
Participants	Participants of the included studies (e.g., students, teachers, health professionals, or institution staff)	Involved stakeholders
Educational level	The participants' educational level (e,g., primary, secondary, and higher education)	Education level
Techniques and technologies	The different AI technologies and techniques within education	The technologies used (e.g., machine learning, deep learning)
Advantages and challenges	The opportunities and challenges brought by AI in education	Opportunities and challenges

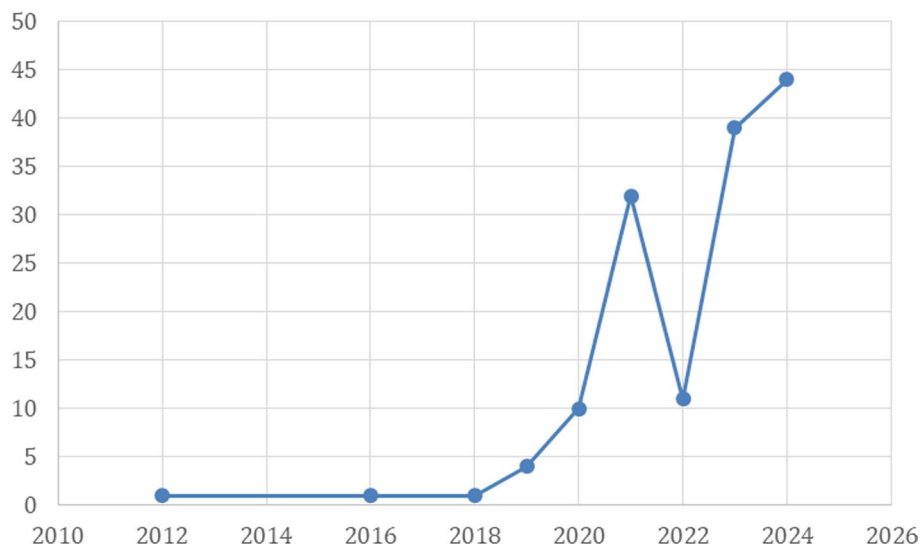

Fig. 3 Distribution of studies by publication year

Results and discussions

The results of the literature review analysis are presented and discussed according to each of the aforementioned research questions.

Trends of AIED literature reviews in terms of publication year, geographic distribution and publication venue

Figure 3 presents the distribution of AIED literature reviews by publication year. The data indicate that the first literature review was published in 2012. This foundational

work, conducted by Drigas and Ioannidou (2012), marked AI as an active research field that had been evolving for over fifty years, emphasizing AI's potential to enhance educational practices, especially for students with special needs through intelligent problem-solving and tailored intervention strategies (Drigas & Ioannidou, 2012).

Building upon this early work, the number of studies examining AI's role in education has increased, driven by significant advancements in AI technology and its growing accessibility (Chen et al., 2020a, 2020b). The number of systematic reviews on AI in education saw a significant uptick starting in 2018, attributed to AI's role as an integral component of modern education systems, replacing traditional methods with learner-centered approaches that cater to individual learning capabilities (Chassignol et al., 2018; Soofi & Ahmed, 2019). This shift is aligned with a broader pedagogical transition towards AI-enabled methodologies that maintain educational goals while enhancing the learning experience and reaching a wider audience (Ferro et al., 2019).

As this trend gained momentum, the utilization of AI in creating adaptive learning environments and providing personalized educational experiences became more prevalent from 2018 onwards, as depicted in Fig. 2. This reflects a growing recognition of AI's crucial role in modernizing education, particularly in the wake of technological advancements that make AI tools more accessible and effective in supporting educational processes (Chassignola et al., 2018).

With the onset of the global pandemic in 2020, the adoption of digital technologies, prominently AI, accelerated as educational institutions sought to adapt to remote learning scenarios. This period witnessed a marked increase in the utilization of AI to support online learning and assessment, as highlighted by Nigam et al. (2021), who extensively reviewed AI-based proctoring systems developed to uphold the integrity of online assessments.

Concurrently, the role of machine learning in precision education has been increasingly recognized, with numerous studies investigating its potential to enhance personalized learning experiences. This surge is attributed to the combined rise of AI capabilities and big data analytics, offering novel approaches to enhancing personalized education (Maghsudi et al., 2021). Additionally, AI has been increasingly utilized to provide personalized support for students, particularly those facing learning challenges (Liang et al., 2021). Luan and Tsai (2021) systematically reviewed 40 empirical studies focused on machine learning applications in precision education, underscoring its capability to predict student performance and engagement in digital learning environments. Since 2022, AI's role in education has remained robust, supporting personalized instruction in K-12 and higher education settings and facilitating innovative learning designs in both academic and professional training environments (Banihashem et al., 2022).

In addition, the application of AI in various educational processes, from content delivery to assessment and feedback mechanisms, has also been a significant focus. Zhai et al., (2021a, 2021b) provided a thorough content analysis of how AI tools, like deep learning and adaptive learning systems are integrated into educational settings, proposing future directions for research including the integration with Internet of Things and swarm intelligence.

Moreover, the ethical, social, and practical challenges of AI in education have become a pivotal area of discourse. Tahiru (2021) analyzed the ethical implications and

challenges of AI adoption in education, emphasizing the need for careful consideration of AI's impact on educational equity and access.

Entering 2024, the landscape of AI in education has reached a pivotal moment, marked by groundbreaking advancements and widespread adoption of AI tools across various educational sectors. This year has seen a significant uptick in the deployment of AI technologies, largely fueled by the maturation of AI systems that are now more capable of handling complex educational tasks with greater efficiency and accuracy (Ogunleye et al., 2024). Major educational platforms have integrated advanced AI functionalities that personalize learning experiences at an unprecedented scale, facilitating real-time adjustments to curricular content based on individual learner profiles and performance metrics (Ogunleye et al., 2024). These developments have not only enhanced the educational landscape but also sparked a substantial increase in academic interest and research into the nuanced impacts of AI on learning outcomes and pedagogical strategies (Alshahrani et al., 2024; Saifudin & Widiyaningtyas, 2024). The focus has particularly been on how AI can bridge gaps in educational access and quality, especially in underserved communities, thereby aligning with global educational equity goals (Alshahrani, 2024). This surge in 2024 has catalyzed a plethora of new studies, exploring both the transformative potential of AI in education and the critical challenges that accompany its integration into traditional educational frameworks.

Analysis of the regional distinctions in the distribution of AIED research, as highlighted in Fig. 4, reveals significant insights into global trends and localized efforts in integrating AI within educational systems. The figure reveals China's leadership with 28 studies, illustrating its extensive engagement with AI to redefine educational paradigms across various learning stages from early childhood to higher educational tiers. The United States and Spain follow, each demonstrating distinctive approaches to incorporating AI into their educational frameworks.

China, with the highest number of studies, demonstrates a profound commitment to embedding AI technologies across various educational sectors. The extensive body of research, as highlighted by Su and Yang (2022) and further expounded by Zhai et al., (2021a, 2021b), showcases a diverse range of AI applications, from early childhood programs to complex tertiary educational systems. This not only reflects China's technological ambition but also underscores a strategic educational framework that integrates AI to enhance personalized learning experiences and address specific pedagogical needs. The focus on precision education through advanced machine learning techniques, discussed by Luan and Tsai (2021), exemplifies China's forward-thinking attempts of leveraging AI to predict and enhance educational outcomes, with the aim to prepare students for a highly digitized future.

In the United States, AI research in education is driven by both necessity and innovation. The significant rise in textbook costs has pivoted the educational focus towards more sustainable digital solutions, such as e-books, catalyzing research into AI's potential to reduce educational expenses while enhancing accessibility and learning outcomes. Studies by Zhang and Aslan (2021), and Sapci and Sapci (2020) illustrate the rapid progression of AI applications within the educational sector, highlighting how AI is increasingly used to create dynamic learning environments that offer personalized learning and real-time assessments. Moreover, the exploration of AI in replacing certain teaching

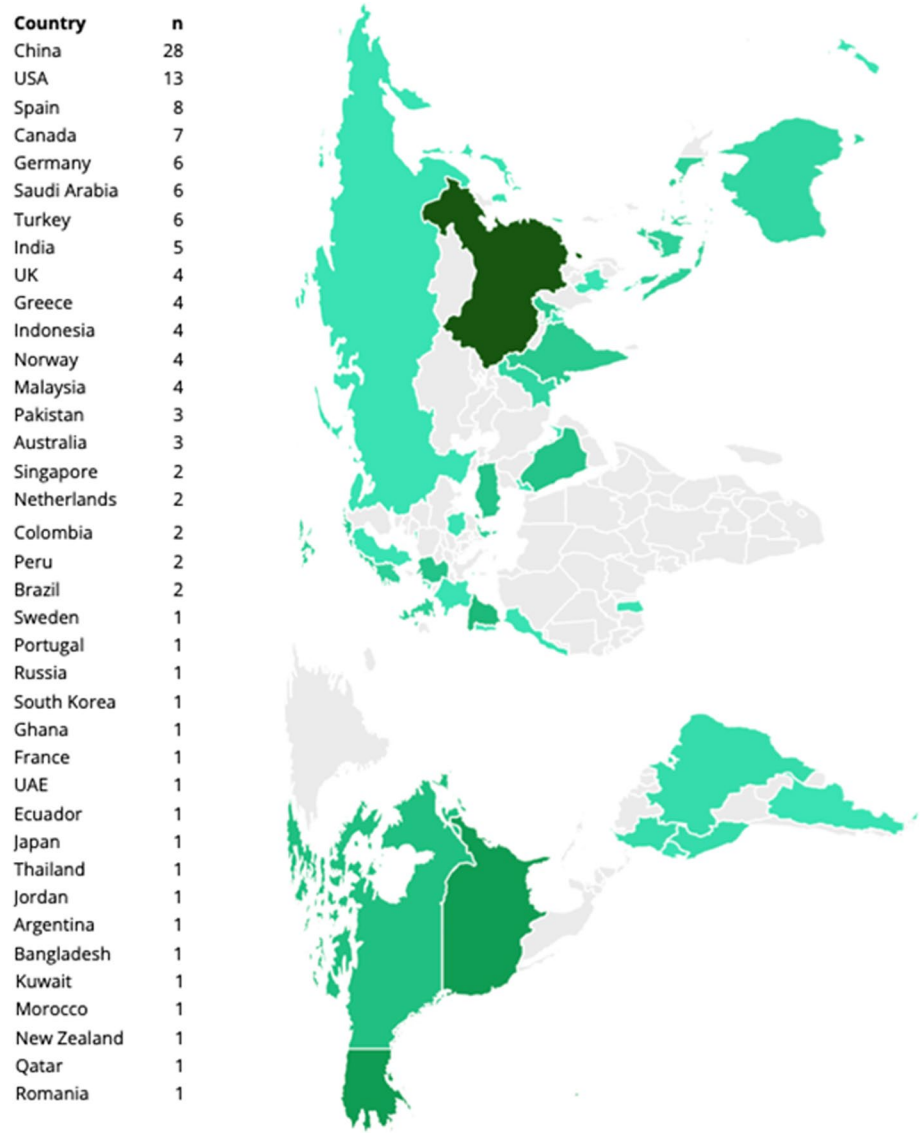

Fig. 4 Distribution of studies by country

roles, as suggested by Edwards and Cheok (2018), indicates a critical response to the ongoing challenges such as teacher shortages, showcasing a pragmatic yet controversial aspect of AI's role in education.

Spain, though with fewer studies compared to China and the USA, focuses on integrating AI within a solid pedagogical framework. The research underscores a commitment to enhancing educational practices through AI while ensuring that technological advancements are aligned with strong educational values and ethics. The work by Vázquez-Cano (2021) highlights this approach, advocating for AI implementations that are not only technologically sound but also pedagogically robust, ensuring that AI tools support and enhance traditional educational methodologies rather than supplanting them.

The regional analysis indicates a diverse yet interconnected global landscape where AI's role in education is both celebrated and scrutinized. Particularly, there is a lack of

research from specific regions like Africa and the Arab region, highlighting the varying priorities and technological readiness across regions that necessitate a collaborative international approach to harness AI's full potential in education effectively. This global perspective not only fosters a deeper understanding of how AI is reshaping educational landscapes across different regions but also highlights the need for continued research and development. Ensuring that AI's integration into education promotes equitable and effective outcomes globally will require ongoing dialogue, policy alignment, and ethical considerations. This will also ensure that AI serves as a tool for educational enhancement rather than as a substitute for fundamental pedagogical principles.

There is currently a lot of interest in the adoption of AIED, as can be seen by the related studies published in 56 peer review journals. In order not to provide a rather long list, Fig. 5 shows only journals having more than one study. This analysis is important because it informs stakeholders about the leading journals in the field and where they can find relevant literature, or where they can publish their own research. Notably, *Computers and Education* emerges as a frontrunner with 23 articles, underscoring a concentrated focus on innovative AI practices within K-12 settings. This substantial contribution reflects the journal's commitment to spearheading in-depth discussions and advancements in AI applications across educational spectrums. In-depth analyses from *Computers and Education* illustrate AI's transformative role across various educational frameworks. Lee and Kwon (2024) emphasize the integration of AI within K-12 education, highlighting project-based learning to enhance AI literacy and ethical understanding among students. This approach advocates for hands-on activities that align with curricular goals, fostering a student-centered learning environment. Concurrently, Li et al. (2024) delve into the learning task design within AI education, stressing the importance of deep conceptual understanding and collaborative educational strategies.

The *IEEE Access* journal follows with 10 articles published. Parallel insights from *IEEE Access* focus on the specialized application of AI through recommender systems within higher education. Chen et al., (2020a, 2020b) discuss how AI has transitioned

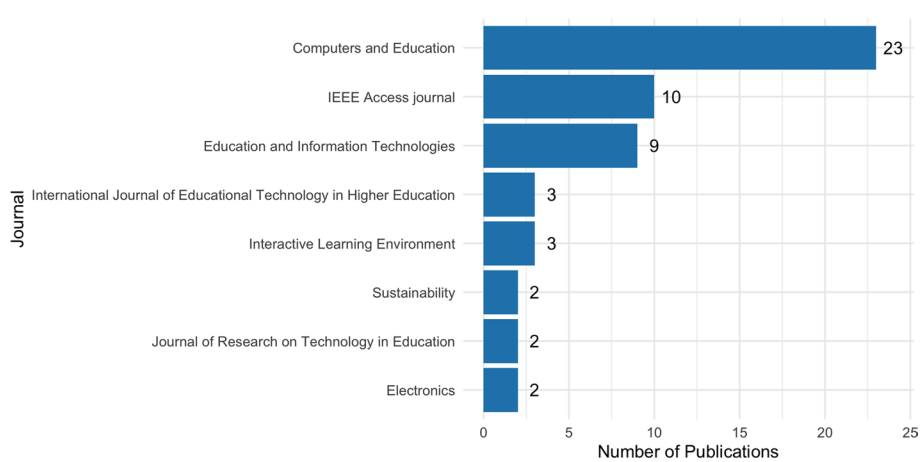

Fig. 5 Distribution of studies by journals

from basic computing solutions to sophisticated systems, like humanoid robots and chatbots, greatly enhancing administrative efficiency and personalized learning in educational settings.

Figure 5 reveals that AIED research has been published in multidisciplinary journals such as *IEEE access*, *Sustainability*, and *Electronics.* This implies that designing and developing AIED requires competencies from different fields. The European Commission (2019) stated that interdisciplinary approaches can help individuals develop a holistic understanding of AI technologies and their potential implications.

AI in education stakeholders and educational levels

Figure 6 presents the frequency and percentages of the AIED stakeholders tackled in the reviewed studies. It is evident that most of the studies (n = 99, 53.51%) focused on students. This might be explained by the fact that students are the most affected by an educational system. Yet, all stakeholders play an important role, as each stakeholder is part of a team that works for the success of educational goals (How & Hung, 2019). Further, teachers (n = 73, 39.45%) were the second-highest AIED stakeholders. It is seen that administrators (n = 10, 5.45%) and researchers (n = 3, 1.62%) had the lowest focus when discussing AIED. It is therefore important that future research studies could investigate, for instance, how AI supports educational leadership and management so that university administrators can make better decisions.

The most neglected stakeholders are found to be special education students and teachers. There are just two review papers (Drigas & Ioannidou, 2012, 2013) that specifically talk about special education. This could be explained with the fact that each disability type has its own features and criteria that should be considered when designing AI-powered technologies in education. This has limited the use of a given AI educational system with all students with disabilities in special education, calling for more personalization and relying on universal design for learning (UDL) to increase accessibility to such systems. Despite the heated controversy over the "no child left behind" motto, the special education section of the AIED review is still ignored. Moreover, most of the AIED review articles targeted students, teachers, or administrators as stakeholders, and neglected other key players that are crucial to the implementation of AIED, such as policymakers.

When analyzing the educational level of the AIED stakeholders in the reviewed studies, Fig. 7 reveals that AIED research predominantly focuses on higher education, accounting for 31% of the reviewed studies. This significant concentration on higher

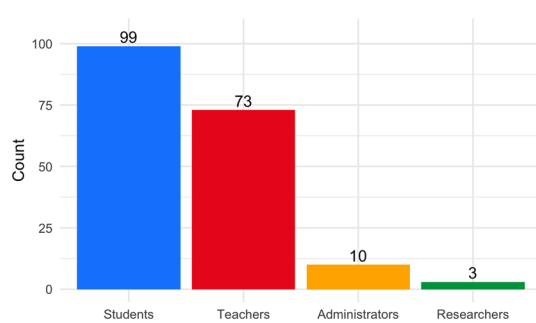

Fig. 6 AIED stakeholders

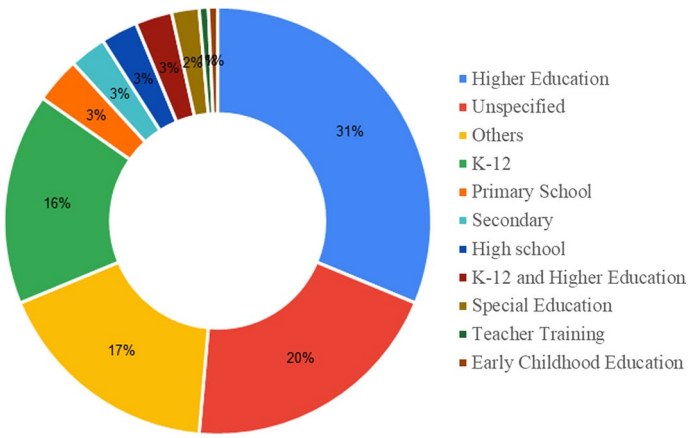

Fig. 7 Distribution of studies by education level

education might be attributed to the widespread implementation of AI-enhanced learning systems within higher education institutions (HEIs). These systems facilitate the development of knowledge and skills through the support of lecturers, tutors, and various technological tools, although the effectiveness of these teaching methods requires further exploration (Kabudi et al., 2021).

Following higher education, K-12 education represents 16% of the studies, reflecting a growing interest in integrating AI technologies within primary and secondary education. However, more investigations are needed to explore how AI is being utilized in these settings, the challenges faced, and the outcomes observed.

Special education, although only representing 2% of the studies, is a critical area where AI's potential to support learners with disabilities is recognized. Studies by Kharbat et al. (2021) and Drigas and Ioannidou (2012) emphasize the role of AI in addressing educational, adaptive, and social skill gaps among students with developmental or intellectual disabilities. These students often face significant health issues that require additional care, leading to academic lags. The integration of AI in their education could mitigate these challenges by providing tailored support and resources.

Interestingly, early childhood education (ECE) is also underrepresented, with only 2% of the studies focusing on this area. However, the potential of AI in ECE is promising, particularly in enhancing learning and development through AI-based toys and smart tools that can improve literacy, inquiry skills, social interactions, and language abilities (Kewalramani et al., 2021; Williams et al., 2019). Further research is needed to explore these applications and their long-term impact on young learners.

Lastly, the use of AI in lifelong learning and professional training, including teacher training, is sparsely covered in the existing literature. Future studies should investigate how AI can be effectively implemented in these areas to support continuous professional development and address the unique needs of learners with mental or physical disabilities, thereby promoting inclusive education.

While the existing research highlights the significant role of AI in higher education, there is a clear need for more comprehensive studies in K-12 education, special education, early childhood education, and lifelong learning. Additional review efforts are

required to deepen the understanding of AI's application and effectiveness across these diverse educational contexts.

Subject area

Figure 8 illustrates the distribution of AIED studies according to the disciplinary categories. It is seen that 56% of the reviewed studies have been in an educational context, whereas they did not specify the disciplines. This might be because several studies took education as a general context, without focusing on a specific subject, to investigate how AI might revolutionize education (Nigam et al., 2021; Tahiru, 2021; Vázquez-Cano, 2021) or also to investigate students and teachers' perception and attitudes towards using AI in education (Zhai et al., 2021a, 2021b).

Figure 8 further reveals that most of the AIED studies have focused particularly on science subjects (16%), such as natural science, mathematics and statistics. For instance, AI has increasingly been employed in science learning and evaluation, especially in the science assessment process, to facilitate automatic scoring efforts (Zhai et al., 2021a, 2021b). This is because STEM education (science, technology, engineering and mathematics) has gained increasing prominence in the last decade. STEM topics are often step-based and well-defined topics where AI tools and techniques can be applied relatively easily (Hwang & Chang, 2023). In addition, mathematics could pose a significant challenge for many students. Therefore, AI could provide an opportunity to diagnose individual students' learning problems and provide personalized support to maximize their learning performance in math.

According to the reviewed studies, the second most use of AIED has been in health and welfare education (12%). For example, AI has been applied in medical education (ME), health informatics, dental study, surgical education and nursing (Ferro et al., 2019; Sapci & Sapci, 2020). One possible reason for the popularity of AI in ME could be that it can provide opportunities for ME to reach larger audiences and integrate learning into active experiences (Ferro et al., 2019). In addition, AI in ME has the potential to generate numerous application possibilities to improve patient care, provide real-time data analytics, and enable continuous patient monitoring (Sapci & Sapci, 2020). Therefore, AI

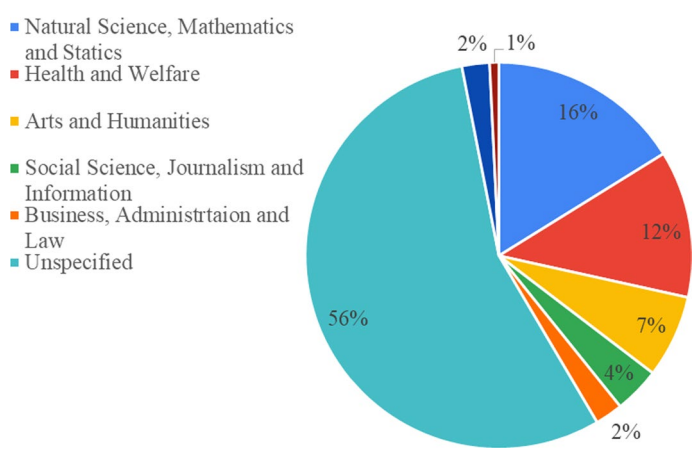

Fig. 8 Distribution of studies by Subject Area

could work as an assistive technology to improve students' access to patient information and reduce medical errors.

Moreover, 7% of reviewed studies focus on implementing AI in arts and humanities. For example, Hahn et al. (2021) mentioned that an automatic feedback system was applied in music education since AI could provide immediate feedback of students, so the instructors could use the feedback to improve students' experience in general musical and instrument learning. Besides, AI application in language education has also grown significantly. Chatbots, Intelligence Tutoring Systems (ITS) and virtual reality systems are frequently used to facilitate language teaching and learning by providing personalized, interactive and authentic language learning context, thus deepening student engagement.

Finally, as shown in Fig. 6, the reviewed studies were less focused on social science, journalism and information (4%), business, administration, and law (2%) and Physical Education (1%). This is not surprising, given the need for more materials and experience in using AI for non-technical subjects (Xu & Babaian, 2021). Hence, more studies should be conducted to explore how to use AI for those less investigated fields.

AIED review method

Figure 9 shows that among all literature reviews, systematic reviews were the most common type of reviews, accounting for 68% of the included articles, followed by meta-analyses at 13%, and bibliometric reviews at 6%. These data suggest that systematic reviews dominate AIED research, indicating that researchers are more focused on systematically summarizing and analyzing existing studies to provide a more comprehensive research perspective and conclusions. For instance, the article by Zhai et al., (2021a, 2021b) conducted a systematic review to analyze the application of AI in the education sector, revealing the current state of AI technology at various educational levels and proposing future research trends and challenges. Similarly, the study by Luan and Tsai (2021) focused on precision education achieved through machine learning, reinforcing the importance of systematic reviews in AIED research.

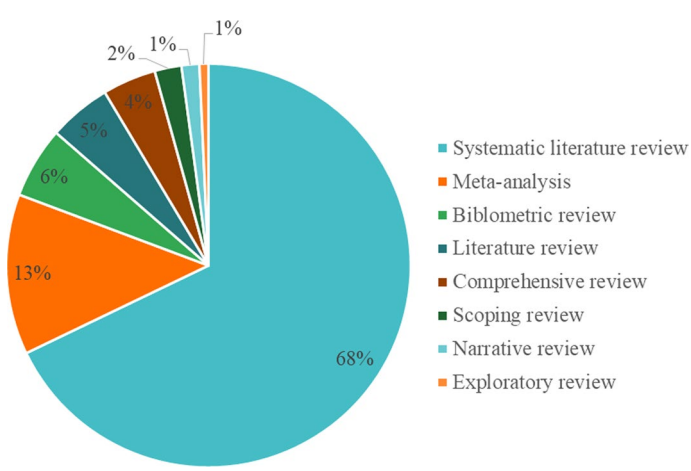

Fig. 9 Distribution by review type

Moreover, while systematic reviews dominate the research trend in AIED, meta-analyses also account for a significant proportion. This type of review quantifies and synthesizes the results of multiple studies, providing more quantitative insights into the application of AI technology in education. The meta-analysis conducted by Zhai et al. (2021a) explored the use of machine learning in science assessment and identified key factors that influence the consistency between machine and human scoring.

The widespread use of systematic reviews and meta-analyses in AIED research demonstrates that researchers are seeking more rigorous and comprehensive methods to evaluate and summarize the effectiveness of AI technologies in education. This also provides direction for future research, indicating that the AIED field needs to continue utilizing systematic reviews and meta-analyses to deepen understanding and advance the field.

AIED objectives

Table 3 presents the frequencies and percentages of the AIED objectives in the reviewed articles. It is evident that the *teaching and learning support* has the highest frequency (n = 95), accounting for 56.5% of the total studies. The analysis reveals that the leading objective of AIED has been mostly to support teaching and learning. The literature refers to it as the integration of various components within the educational process, where an educator identifies and creates the learning objectives, develops teaching resources, and implements the teaching and learning strategies. Teaching and learning support also involves decreasing the workload of teachers and students, thereby, making the learning process more efficient and effective. For instance, AI could be used to provide pedagogical or instructional support (Hannan & Liu, 2021; Xu & Ouyang, 2022) by automating manual tasks, since, thanks to the internet, teaching and learning resources can be accessed from anywhere, at any time.

Also, frequently asked questions can be answered through chatbots and AI tutors (Kanglang, 2021; Vázquez-Cano, 2021). AI could also support students in learning at their own pace, improve their quality of learning, and even support students with intellectual disabilities from education and health perspectives (Kharbat et al., 2021). However, the analysis demonstrates that, since teaching and learning are AIED's primary areas of attention, teaching and learning support has been the subject of the majority of publications that were reviewed. Another meaning of teaching and learning support is to provide a variety of teaching and learning activities (Costa et al.,

Table 3 AI in education objectives

AI in education objectives	Frequency	Percentage
Teaching and learning support	95	56.5%
Teaching and learning assessment	24	14.3%
Adaptive teaching and learning	21	12.5%
Student prediction	13	7.7%
Teacher and student profiling/modelling	8	4.8%
Monitoring students	7	4.2%

2017), such as student activities (e.g., procedural math problems), and AI could support the whole spectrum (Chan, 2023), including social emotional learning (Sethi et al., 2024), such as enabling learners to be active in small group, and collaborative learning (Kim et al., 2022).

The second most often mentioned objective in the review articles is *teaching and learning assessment* (n = 24, 14.3%). Teaching and learning assessment are categorized in two types: formative or summative assessment of students (Chassignol et al., 2018; Drigas & Ioannidou, 2012; González-Calatayud et al., 2021; Zhang & Aslan, 2021). The high amount of discussion around teaching and learning assessment may have been catalyzed by COVID-19, where assessment was one of the key issues in online teaching and learning processes (Asgarova et al., 2021; Zulaiha et al., 2020). Ouyang and Jiao (2021) provide three types of teaching and learning assessment. The first one is where the learner is the recipient, and the AI directs cognitive learning. The second one is where the learner is collaborating with AI for learning and AI supports cognitive learning. The third type is where the learner directs cognitive learning and AI is used to empower the learner.

The third AIED objective is found to be *adaptive teaching and learning* (n = 21, 12.5%). It is an approach aimed at achieving a common instructional goal with learners considering their individual differences (Ikwumelu et al., 2015). Borich (2011) defined adaptive teaching and learning as applying different instructional strategies to different groups of learners so that natural diversity prevailing in the classroom does not prevent any learner from achieving success. Daugherty et al. (2022) presented adaptive learning and teaching as an individualized adaptation of content and pedagogy implementation according to the needs of the participants to increase the effectiveness and quality of learning.

AI could enhance adaptive teaching and learning when using personalized learning platforms that adapt to students' learning strategies, the sequence and difficulty of the task abilities, the time of feedback and students' preferences (Pliakos et al., 2019). At the same time, AI adaptive teaching and learning systems can provide timely and targeted interventions, ensuring that learners receive the most relevant and effective educational materials and activities.

The least discussed AIED objective is related to monitoring students (n = 7, 4.2%), calling for more investigation in this regard. Vilanova et al. (2019) defined student monitoring as a practice that helps teachers use student performance to continually evaluate the effectiveness of their teaching and make more informed instructional decisions. Monitoring students also refers to tracking their behavior, academic performance, and engagement in real-time. AI can enhance this process by analyzing data to identify patterns, predict outcomes, and provide personalized feedback to improve learning outcomes. However, there is a need to design AIED applications to support teachers in monitoring students to facilitate teaching and learning. The design of AIED applications could face challenges such as AI-based misunderstanding, misleadingness, limitations, the weak connection to theoretical pedagogical perspectives, and the need for deep exploration of ethical and educational approaches in the application of AIED in education (Akgun & Greenhow, 2021; Zawacki-Richter et al., 2019).

Table 4 AI technologies/techniques in education

AI techniques	Frequency	Percentage
Machine learning (ML)	50	23.80%
Large language models (LLMs)/generative AI	41	19.50%
Intelligent tutoring system (ITS)	27	12.85%
Chabot	20	9.50%
Deep learning (DL)	10	4.76%
Augmented/virtual reality (AR/VR)	10	4.76%
Robotics	9	4.28%
Adaptive systems (AS)	8	3.80%
Recommender system	8	3.80%
Natural language processing (NLP)	7	3.30%
Decision support system	7	3.30%
Exam monitoring systems	4	1.90%
Fuzzy logic	3	1.42%
Support vector machines	3	1.42%
Knowledge Tracing	1	0.47%

AIED techniques/technologies

There are several techniques and technologies that are associated with the use of AI and this can differ between sectors and use cases (Das et al., 2015). For example, studies have highlighted the different AI techniques and technologies that can be applied in various domains including medicine (He et al., 2019), industry (Ahmed et al., 2022), and energy (Ali & Choi, 2020). When focusing on the field of education, the most commonly used AI techniques and technologies are displayed in Table 4. Specifically, machine learning emerged as the most widely used technique (23.8%) with deep learning technique also being the focus of some studies (4.76%). These outcomes are in line with those of Kučak et al. (2018) and further highlight the significant role that machine learning can play in assisting teachers, students, and administrators as it can support teaching and learning activities through the enrichment of assessment and monitoring processes, the provision of personalized recommendations, and the prediction of students' performance and learning needs (Luan & Tsai, 2021; Rastrollo et al., 2020; Sanusi et al., 2023). Similarly, deep learning approaches are also being used in the context of educational data mining (Hernández-Blanco et al., 2019) across educational levels (Winje & Løndal, 2020) to enrich the educational process through the provision of personalized material, recommendations, tasks, and feedback (Perrotta & Selwyn, 2020).

Despite the recent developments of Large Language Models (LLMs) and Generative AI, the focus on implementing them in education has rapidly increased. Specifically, 19.5% of the studies focus on the use of LLMs and generative AI in educational settings. LLMs refer to large and complex computational models that have been trained on vast data amounts and are capable of performing NLP tasks, such as comprehending and generating different content types (Wei et al., 2022; Zhao et al., 2023). Given the recency of their wider introduction to educational settings, with 2023 being the main publication year for documents associated with the use of LLMs and education, and the predominant focus on the use of ChatGPT in relation to other models, several

studies emphasize (Dong et al., 2024; Tlili et al., 2023a) the need to further examine their role, integration in, and impact on teaching and learning activities.

9.5% of the studies focused on integrating chatbots. Chatbots are computer programs that simulate human conversations by providing diverse information and knowledge types and promote interpersonal communication through easy-to-use and intuitive interfaces (Adamopoulou & Moussiades, 2020; Hwang & Chang, 2023). Recent studies have highlighted the opportunities that these techniques can bring in teaching and learning but also noted the challenges and the barriers that need to be addressed (Kasneci et al., 2023; Meyer et al., 2023). However, since the integration of these techniques can affect both students and teachers, emphasis should be placed not only on exploring how students use them in their learning but also how teachers utilize them to plan and carry out their lessons (Jeon & Lee, 2023).

Intelligent tutoring systems (ITS) are also being commonly used (12.85%) in the context of AIED. Specifically, ITS refer to computer-assisted learning environments that can be applied in various educational levels and subjects that utilize computational models to offer student-centered, personalized, and self-paced learning opportunities to help students acquire knowledge (Graesser et al., 2012; Ma et al., 2014) by tailoring the instructional methods, approaches, and activities according to each individual learner (Mousavinasab et al., 2021). Additionally, ITS take learners' unique traits, knowledge, characteristics, preferences, and skills into account to provide them with adaptive and personalized learning (Anderson et al., 1985) as they are capable of identifying, analyzing, and monitoring learners' psychological states and dynamically adapting to their needs (Graesser et al., 2012).

Hence, the computerized learning environments offered by ITS include various knowledge types, models, and components, use complicated computational models, and are based on learners' cognitive and motivational states (Conati, 2009; Lampropoulos, 2023a; Xu et al., 2019). When combining ITS with extended reality technologies (e.g., augmented reality (AR), virtual reality (VR), and mixed reality (MR)), which are another category of technologies on which emphasis is being placed in the context of AIED (4.76%), immersive and interactive ITS that can positively affect learning outcomes can be created (Lampropoulos, 2023a).

Despite the various technologies and techniques in AIED, the effectiveness of AIED on learning achievement remains questionable (Rodway & Schepman, 2023). This calls for more investigation on how teachers and students could work with AI to achieve a given educational objective and how AI could augment them rather than replacing them. This further emphasizes the importance of collaborative intelligence and how artificial intelligence and human intelligence could be combined together to achieve a given educational objective. Therefore, future research directions should consider this line of research investigating human–machine collaboration and human–machine teaming up in education.

AIED advantages and challenges

The analysis of AIED literature reviews has identified significant advantages that AIED can bring to education. At the same time, there are various challenges that need the

Table 5 Advantages of AI in education

Positive effects of AI on education	Frequency	Percentage
Students' and teachers' assessment support	38	21.50%
Provide personalized learning	31	16.66%
Improve teaching and learning quality	30	16.12%
Understand the learning processes	16	8.60%
Enhance learning outcomes	15	8.06%
Increase students' cognitive engagement and motivation	14	7.52%
Improve students-teachers—system-administration interaction	13	6.98%
Improve creativity and critical thinking	7	3.76%
Improve content accessibility	6	3.22%
Promote AI literacy	5	2.68%
Decrease work stress	4	2.15%
Emotional support	2	1.07%
Improve exam monitoring	1	0.53%

attention of researchers and practitioners. The following sections discuss these advantages and challenges.

AIED advantages

Table 5 presents the frequencies and percentages of the AIED advantages mentioned within the reviewed papers. The most frequently mentioned advantage is providing personalized learning (n = 31, 16.66%). Personalized learning is a learner-centered approach that tailors individual students' needs by providing them with customized content, pace, and methods (Essa et al., 2023). AI is being used to provide personalized learning by using adaptive systems and learning analytics to model students and then provide adequate learning paths according to their profiles (Raj & Renumol, 2022). Such an approach allows for providing real-time personalized support to ensure that learners receive the most relevant and effective instructions.

Furthermore, most of the studies have highlighted the improvement of the teaching and learning quality (n = 30, 16.12%) as another benefit of integrating AI in educational settings. Specifically, studies have showcased the potential of AI to improve teachers' and students' skills, knowledge, critical thinking and creativity (Alfaro et al., 2020; Zhai et al., 2021a, 2021b) as well as the overall learning experience and quality through the provision of adaptive and personalized learning which takes students' unique traits, needs, goals, and existing skills and knowledge into account (Ayeni et al., 2024; Kabudi et al., 2021; Pratama et al., 2023). Another benefit that can be yielded through the use of AI in education is the improvement of assessment and support systems for students and teachers (n = 28, 15.05%). Specifically, through its potentials to provide users with personalized feedback and recommendations, AI can enrich and transform assessment and support systems in educational settings by offering them tailored activities and material as well as by automating everyday tasks (González-Calatayud et al., 2021; Martínez-Comesanã et al., 2023).

Moreover, several benefits in terms of learning performance were observed. Particularly, AI has emerged as a meaningful educational tool that can improve students' understanding of the learning process and subject (n = 16, 8.60%), increase their learning

outcomes and performance (n = 15, 8.06%), enhance their cognitive engagement and motivation (n = 14, 7.52%), improve the interactions among education stakeholders (n = 13, 6.98%), and increase their critical thinking and creativity (n = 7, 3.76%). Among the benefits that emerged, decreased work stress (n = 4, 2.15%), emotional support (n = 2, 1.07%), and exam monitoring (n = 1, 0.53%) were observed in fewer studies. The need for exam monitoring, particularly, has further increased especially during the COVID-19, where several universities reported challenges on how to design online exams and monitor students to prevent cheating. Therefore, more research is needed in this regard.

AIED challenges

However, to reap the benefits of AI in education, there are several challenges that should be addressed. Table 6 presents the most common challenges identified. The challenges were grouped into several categories, namely ethical and safety (n = 49), AI technologies (n = 34), design (n = 28), teachers (n = 21), and students (n = 10). In the ethical and safety category, data privacy (n = 29, 20.42%) emerged as the main concern and challenge that needs to be addressed, followed by the need to ensure inclusion and equality (n = 8, 5.63%) as well as high quality and reliability (n = 6, 4.22%). Specifically, the collection, processing, and use of personal data is an important aspect that should be further examined, since confidential information of education stakeholders is at stake (Chen et al., 2020a; Nigam et al., 2021; Zawacki-Richter et al., 2019). For AI to be more widely used in educational settings, it is important to examine how to develop, introduce, and use AI systems that foster an inclusive and equitable education (Hannan & Liu, 2021; Xia et al., 2022) that empowers all education stakeholders.

Table 6 AIED challenges

Category	Challenges	N	%
AI technologies	Lack of humanity	11	7.74%
	Limitation of AI use	5	3.52%
	Lack of teaching and learning support	4	2.81%
	Lack of personalization	10	7.04%
	Cost	4	2.81%
Teachers	Lack of AI training	14	9.85%
	Changes in teaching methods	7	4.92%
Students	AI digital literacy	7	4.92%
	Reduce students' critical thinking	1	0.70%
	Cheating	1	0.70%
	Plagiarism	1	0.70%
Design	Difficulty in application	14	9.85%
	Difficulty in system development	14	9.85%
Ethical and safety	Data privacy	29	20.42%
	Inclusion and equality	8	5.63%
	Quality and reliability	6	4.22%
	Legal responsibility	2	1.40%
	Intellectual property	1	0.70%
	Liability	1	0.70%
	Social norms	1	0.70%

Additionally, as AI systems are being used to personalize learning by recommending appropriate educational resources and tasks, and to improve monitoring and assessment algorithmic bias in AI systems, it is an important topic that should be further examined as it can significantly impact the outcomes (Ntoutsi et al., 2020; Yang et al., 2021a). The ethical issues of using AI in education remain an open challenge for both researchers and educational practitioners (Zhai et al., 2021a, 2021b) and there is a clear need to formulate suitable policies, standards, guidelines, and frameworks (Lee et al., 2021; Tahiru, 2021; Zawacki-Richter et al., 2019) and to develop AI literacy training programs that emphasize ethical awareness and social responsibility in using AI (Ng et al., 2021).

Lack of humanity (n = 11, 7.74%) and of personalization (n = 10, 7.04%) emerged as the main challenges within the AI technologies category. The limitations of AI use (n = 5, 3.52%), the lack of teaching and learning support (n = 4, 2.81%), and the cost associated with the integration of AI in education (n = 4, 2.81%) were additional challenges identified but reported in fewer studies. These outcomes highlight the fact that machines are far from human intelligence and emotions and cannot undertake the role of teachers (González-Calatayud et al., 2021; Zhang & Aslan, 2021). Additionally, although AI systems can enrich the evaluation and assessment systems, they do not always result in correct predictions, decisions, or course of action (Chassignol et al., 2018); thus, highlighting the need for teacher intervention and supervision. Additionally, to provide meaningful personalized learning, there are challenges associated with the identification of students' characteristics, preferences, knowledge, and skills that should be addressed (Guan et al., 2020). This is particularly true in the case of students with intellectual and developmental disabilities in which identifying the strengths, needs, capabilities and providing adequate guidance is more difficult. Learning analytics techniques are often used in the context of AI intervention to further improve personalization and adaptability which are essential elements in the wider adoption of AI in education (Chen et al., 2020a, 2020b; Lampropoulos, 2023b).

Furthermore, challenges were also observed in the design and development of AI systems (n = 14, 9.85%) as well as their application in teaching and training settings (n = 14, 9.85%). The lack of research into the identification of the most appropriate methods and strategies in developing adaptive and personalized educational AI interventions hinders a wider adoption of AI, which in combination with the lack of teachers' AI and digital literacy (González-Calatayud et al., 2021) in creating effective AI-enabled educational material and activities further limits its use in education (Du et al., 2024; Hwang & Tu, 2021; Ng et al., 2021). The complexity of developing effective AI systems, the technical skills associated with it (Hwang & Tu, 2021; Kharbat et al., 2021) and the fact that in many cases, these systems are "black-boxes" that do not describe the specifications of the model creation are additional reasons that challenge the use of AI in education (Zhai et al., 2021a, 2021b).

The challenges identified are also associated with the use of AI by teachers and students. Specifically, teachers are faced with a lack of AI training (n = 14, 9.85%) in designing, using, and integrating AI in their classrooms as well as with a drastic change in their teaching methods and approaches (n = 7, 4.92%). The lack of appropriate AI training hinders teachers' ability to create effective and meaningful learning and to apply appropriate learning approaches which also affects their students' learning (González-Calatayud

et al., 2021; Ng et al., 2021). These outcomes directly affect students, who are lacking in AI literacy, since in addition to keeping up with the curriculum, they also have to familiarize themselves with different AI tools (Ng et al., 2021; Xu & Ouyang, 2022).

Conclusions and research agenda

The field of AI is advancing rapidly and is integrated into different domains of research and practice every day. Due to the potential benefits and the ability to transform teaching and learning, AI has become increasingly discussed, adopted and researched. Such rapid increase has been obvious in the multitude of studies being carried out worldwide. Therefore, several literature reviews have been conducted to map the existing literature from different perspectives. The fragmented and varied landscape and scope of existing literature reviews motivated us to perform an overarching systematic review that synthesizes all existing systematic reviews to offer a grand perspective of the state of the art. We further enhance our review with a future research agenda based on our analysis and synthesis of existing studies. To do so, we examined 143 literature reviews from 2012 to 2024 covering a wide geographic distribution, multiple venues, research traditions and educational levels. This comprehensive coverage allows us to have a broad perspective as well as to devise a research agenda in form of recommendations, challenges and research directions.

Current status of the research on AI

First and foremost, there is a surge of interest in AIED across all stakeholders and in particular during the last two years, 2023 and 2024. Of course, this surge is due to the recent emergence of generative AI being more widely accessible to both individuals and consumers. More importantly, because generative AI was able to replicate some of the high cognitive functions that used to be an exclusive preserve of highly educated individuals. In doing so, AI has demonstrated its capability to be a more credible agent in the learning process and subsequently a possible viable alternative to humans– at least in theory so far.

Researchers from all over the world have contributed —or shall we say rushed— to the exploration of AI to study its potentials, and harness the promised benefits. Most of the studies reviewed here focused on the use of ML, LLMs and Generative AI at large

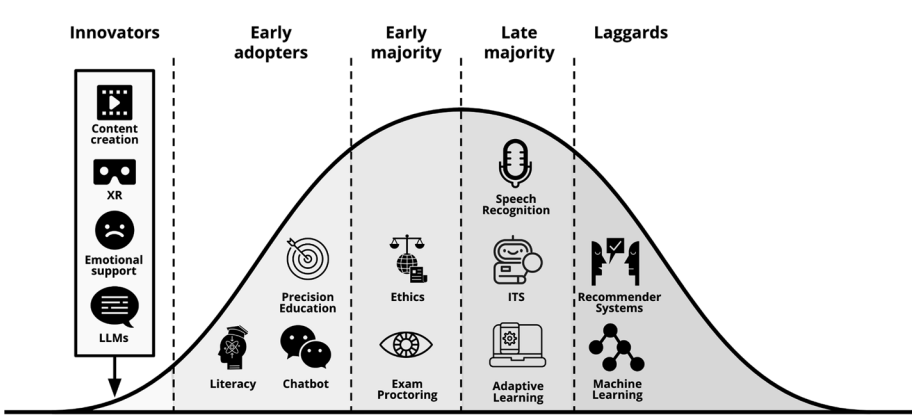

Fig. 10 AIED adoption curve

as well as ITS, and Chatbots. A prime focus was placed on how AI can support teaching and learning, how it can be integrated into assessment and monitoring tasks, and how it can be used for prediction, profiling, and modeling to offer adaptive and personalized teaching and learning (see Fig. 10 for a summary). An overarching conclusion is that the effectiveness of AIED to improve learning outcomes remains far from conclusive, especially in the long-term. As we currently stand, most studies are explorative, short-term and in limited domains e.g., language and academic writing. Therefore, it remains to be seen how AI would prove effective across the wide gamut of skills and competencies that are embedded across the diverse educational ecosystem.

Figure 10 depicts the AIED adoption curve based on a qualitative mapping of the prevalence of AIED tools and applications (see Table 4) and their temporal trends from the analyzed systematic reviews to the five stages of the technology adoption model. At the top of the adoption curve, in the late majority and laggard's phases, we see AI technologies that have achieved widespread use. Recommender systems and machine learning models, once considered groundbreaking, are now embedded in a vast array of applications and educational technologies to personalize content delivery and offer custom recommendations. Alongside these are speech recognition technologies, which have been integrated into learning platforms to assist with accessibility, such as dictation tools, automated transcription services and language learning. Closely related are ITS which has been around for while offering real-time feedback and personalized content. The adoption of such technologies has been associated with increasing emphasis on ethics and privacy. In the innovation phase, where lower adoption is rapidly changing, we see AI literacy, precision AI and the emerging conversation agents. At the innovation stage, the experimental and cutting-edge technologies lie where we see extended reality (XR), emotional support systems, and large language models (LLMs), which promises to shape the future of AI in education. As such, we see an AI adoption curve moving from tools that enhance basic educational processes, like recommender systems and adaptive learning, to more advanced, exploratory technologies that are more embedded in the learning process and human cognitive process.

A prominent gap that we observed is a lack of focus on childhood, primary and secondary education, as well as on teacher training and special education. AI has become in the hands of young children and it is so far unknown how modern AI tools would affect the cognitive development of children, their learning behavior and their learning skills. This fact highlights the need to consider all education stakeholders' and devote more resources to how AI would influence the development of students at the young side of the spectrum.

AI as a technology

Regarding AI as a core technology, several challenges remain unresolved (e.g., cost, use limitations, lack of humanity, personalization, and support), to teachers (e.g., lack of appropriate AI training, changes in teaching methods), to students (e.g., AI digital literacy, cheating and plagiarism, reduced critical thinking), to design (e.g., difficulty in application and in system development), and to ethical and safety concerns (e.g., data privacy, equality and inclusion, reliability and quality, legal responsibility, copyright infringement and lack of consent etc.).

Bias and cultural diversity

AI comes with a huge risk of bias, although some effort is being done to address it, it is so far lagging and largely ineffective. Furthermore, the current approach to AI aggregates data from many others to devise recommendations. The training data reflects the average behavior, the WEIRD (Western, Educated, Industrialized, Rich, and Democratic) traditions or simply, the available data that made it to the internet (Kilanioti et al., 2024). In doing so, AI models magnify the flaws and limitations inherent in our current world essentially creating a one model that is expected to work for all. Given that diverse training data is far from easy to obtain, we may see more echo-chambers and amplification of biases. Therefore, serious efforts need to be made to individualize AI recommendations, considering the biases that are hard wired into the way AI models are built and more importantly, humanize AI and invest in AI literacy (Saqr et al., 2024). Furthermore, more research is needed to assess the bias in recommendations, suggestions and different outputs in AI as well as methods to mitigate such biases (Varsha, 2023), particularly those arising from non-representative training data (Norori et al., 2021). This includes creating frameworks for individualized AI recommendations that account for diverse cultural, socio-economic, and demographic contexts.

AI literacy

Within the current conversation about AI, there is little discussion about AI literacy and in particular, critical AI literacy. We need to prioritize a responsible approach that fosters critical AI literacy among all stakeholders and the society at large (González-Calatayud et al., 2021). This involves empowering stakeholders to deeply understand, evaluate, and question AI systems and tools, especially regarding their safety, biases, and decision-making processes. In doing so, we can promote the development of AI solutions that prioritize equality, fairness, and inclusivity in education (Ayeni et al., 2024; Hannan & Liu, 2021; Xia et al., 2022). This needs to be examined across AI training programs, curricula, and policies (Ayeni et al., 2024). Future research should explore effective strategies for integrating critical AI literacy into education curricula and professional training programs. Studies should investigate the long-term effects of critical AI literacy on promoting fairness, equality, and inclusivity within various educational and professional settings. Furthermore, interdisciplinary studies that integrate insights from social sciences, ethics, and AI technology are crucial to humanize AI and enhance AI literacy among users and developers alike (Ng et al., 2021).

Alignment

Equally important is the faint presence of research on alignment in education. Alignment is the process of ensuring that AI is integrated in a way that supports the human values and objectives to ensure that AI tools and applications are consistent with human educational principles, student needs, and ethical values (Kasirzadeh & Gabriel, 2023). Alignment aims to make sure that efficiency and financial gains are not the main driver of AI but rather the developmental, cognitive, and emotional needs of learners. Proper alignment would also ensure that AI supports sane pedagogical principles that enable creativity, critical thinking, and problem-solving (Kim et al., 2022). Successful alignment

means that AI meets student needs, promotes personalized and inclusive learning experiences that accommodates diverse backgrounds, abilities and adjusts to individual differences (Ayeni et al., 2024). Without alignment, we risk an AI that prioritizes efficiency over meaningful educational outcomes.

Academic and cognitive development

The wide availability of AI has increased the risk that students become overly dependent on AI for tasks like essay writing, problem-solving, or coding (Zhai et al., 2024). Such a reliance could interfere with the natural development of critical thinking, creativity, and problem-solving skills as well as long-term retention of knowledge (Dong et al., 2024). As such, the emergence of AI brings new challenges to how humans learn, work or interact with the world. The rapid pace of AI evolution raises questions about the influence of AI on the skills needed for learning and whether institutions are working to equip students with such skills. All the more so, we have little knowledge of which of such AI skills—or AI uses—are conducive to learning or may be endangering basic human skills such as problem solving or critical thinking. Therefore, research is needed to study the impact of AI on students' behavior, cognition and learning skills as well as prepare students for a future where AI skills could be at the forefront of the professional workplace (Seo et al., 2021).

Future research should study how to design the integration of AI in education in a way that strikes a balance between AI and human skill development. More importantly, researchers should investigate the long-term effects of AI on students' cognitive development, creativity, and problem-solving abilities (Albadarin et al., 2024), as well as identify and develop effective pedagogical approaches that balance AI utilization with the development of essential human competencies (Dong et al., 2024; Hadi Mogavi et al., 2024).

AI human interaction and collaboration

We are facing a new reality where students and AI interact, work together and collaborate —and at times may conflict. How students interact with AI, regulate such interactions, and how such interactions impact students are an open area of inquiry. Research needs to understand how such interactions emerge, evolve or in which way it impacts the socio-cognitive development of students. Therefore, research is needed to assess human-AI teaming up and co-evolution, co-intelligence, hybrid intelligence or collaborative intelligence (i.e., combining human intelligence and artificial intelligence) (Järvela et al., 2023).

Future research should explore the dynamics of human-AI collaboration in educational settings, and assess how these interactions emerge, evolve, and impact students' socio-cognitive development. This includes examining the regulatory mechanisms students use to manage their collaboration with AI and the long-term effects on their cognitive, emotional, and social skills (Järvela et al., 2023). Furthermore, researchers should investigate best practices for optimizing human-AI teaming that ensures the benefits of AI integration are maximized while potential conflicts and negative effects are effectively mitigated.

AI and academic integrity and assessment

AI may exacerbate academic dishonesty issues by making it easier for students to present AI-generated content as their own work. This practice not only undermines the value of honest academic efforts but also challenges educators in assessing true student performance. The prevalence of such plagiarism (Albadarin et al., 2024) devalues authentic learning experiences and erodes the trust essential for a credible educational environment. Researchers have been actively researching this topic as well as institutions actively looking for solutions. However, current plagiarism detection methods have been vastly unreliable which calls for new methods of assessment.

Future research should find innovative assessment techniques that emphasize critical thinking, problem-solving (González-Calatayud et al., 2021), and reduce reliance on traditional assessment modalities which are vulnerable to AI-assisted plagiarism (Albadarin et al., 2024). Researchers could also investigate the development of AI detection tools tailored to evolving AI-generated content and examine how educational systems might balance technological advancements with ethical considerations.

True and long-term impact

Given the recency of AI, long term impact on students' skills is largely unknown and it would take time before we figure out how AI impacts human learning, behavior or skills. Several papers have reported short term gains in, e.g., knowledge. However, how long such gains remain is yet to be determined. Therefore, we urge researchers and practitioners to be cautious and not to confuse short term gains with actual long-term gains that are transformative with real impact.

Among the challenges that are absent from educational research discourse is how we judge the hype about AI and the byproducts thereof. Over the years, AI has been plagued by perpetual cycles of overhype in which the research community has gone through several disappointing cycles of hype, disillusionment, wide scale criticism followed by dryness of funding. Therefore, we have to be realistic in approaching the results and potentials of AI. It is imperative that rigorous research is more needed than ever before so that researchers do not amplify or confuse small gains—or actually the lack thereof—with true gains that are conducive to deep learning or transformative to education.

A large number of emerging research is testing questions related to AI generating content (e.g., code, assessments or images). A deluge of papers is just "AI can generate something" or "AI can do something". However, empirically assessing the power of generative AI to generate content is of little importance to the learning processes if it does not assess the impact of such materials on learning, use a theory informed approach or help update our theoretical understanding of learning and teaching. All the more so, we need to have a deeper look about how such generated content can actually support students' learning with a deeper understanding of the long-term outcomes. For instance, the fact that LLMs can generate programming code is of little value for students learning programming. Nevertheless, how such generated code can support critical thinking, problem solving and computational thinking skills is what matters to students' learning. Moreover, how using LLMs can impact programming education, skills, and our approach to the subject matter is far more important.

AI and learning design

In the same vein, the impact of AI on the learning design and education technologies may be transformative in that many of the existing tools would include AI elements. Some existing tools may become obsolete (e.g., writing tools) and others may change drastically (e.g., automatic assessment tools). Therefore, learning design is at a major turning point. Therefore, appropriate design considerations and methodologies should be adopted for each use case and suitable policies should be developed.

AI and the environment

There is a lack of serious conversation about the energy footprint of AI and how such systems can rapidly consume massive amounts of energy or add to the already troubled environment. In the same token, less is known or discussed about the fair labor practices in data preparation. While these topics may seem far from education, they are relevant on many levels given that learners are consumers, and tomorrows' leaders who will shape these technologies. Research should therefore focus on sustainable AI practices and responsible AI literacy. Integrating these considerations ensures that the adoption of AI in education aligns with broader societal responsibilities toward the environment and fair labor practices.

Concluding remarks

Integration of AI in education is still in its early stages and therefore, there is a need for more studies that explore the adoption and use of AI in education to be conducted. Future studies should focus on identifying the most suitable approaches and methodologies to effectively introduce, adopt, and use AI in education as well as AI capabilities to support and meet the needs of different education stakeholders. Emphasis should also be placed on examining how education stakeholders' characteristics, knowledge, and background affect their use of and viewpoints regarding AI. Additionally, there is a need to further examine how to increase education stakeholders' AI literacy and how to provide educators with effective training programs. Further studies should also examine the use of AI in different educational levels and subjects and put emphasis on its use in special education (Drigas & Ioannidou, 2012, 2013) and on increasing inclusivity and personalization (Maghsudi et al., 2021).

Last but not least, the most important critical challenge we face today is managing the transition to a world where AI is prevalent. A well-managed transition that equips students with the needed skills, protects them from the possible harms and more importantly promotes diversity and inclusion. A transition where benefits are shared widely by the society at large without bias. A transition that avoids the paradox of plenty, in which the society is rich in aggregate and many are left behind in need. While this study provides solid ground on future research directions related to AIED, it still has some limitations that should be acknowledged. First, the findings of this study are limited by the searched keywords and included databases. Additionally, the obtained results are drawn based on the covered journal papers. However, several relevant conferences were not covered, such as the International Conference on

Artificial Intelligence in Education (AIED). Therefore, future studies could complement the present study by covering these conferences.

Abbreviations

AI	artificial intelligence
AIED	artificial intelligence in education
ECE	early childhood education
ME	medical education
ITS	intelligent tutoring systems
ML	machine learning
NLP	natural language processing
LLMs	large language models
VR/AR/XR	virtual reality/augmented reality/extended reality (XR)

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Availability of data and materials
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Declarations

Competing interests
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